



University of Asia Pacific

Admit Card

Final-Term Examination of Fall, 2025

Financial Clearance

PAID

Registration No : 22201243

Student Name : H. M. Tahsin Sheikh

Program : Bachelor of Science in Computer Science and Engineering



SI.NO.	COURSE CODE	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
1	CSE 400-A	Final Year Design Project	3.00	
2	CSE 401	Operating System	3.00	
3	CSE 402	Operating System Lab	1.50	
4	CSE 405	Numerical and Mathematical Analysis for Engineer	3.00	
5	IMG 401	Industry and Operational Management	3.00	
6	CSE 413	Machine Learning and Deep Learning	3.00	
7	CSE 437	Robotics	3.00	
8	CSE 414	Machine Learning and Deep Learning Lab	1.50	
9	CSE 438	Robotics Lab	1.50	

Total Credit: 22.50

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.

2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.

3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.

4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall. Violators will be subjects to disciplinary action.

This is a system generated Admit Card. No signature is required.

Admit Card Generation Time: 25-May-2026 11:32 PM

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 2025
Program: B.Sc. in CSE
4th year 1st Semester

Course Title: Operating Systems
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 401
Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. A multinational banking organization maintains customer transaction records on a magnetic disk storage system. During peak transaction hours, the operating system must efficiently schedule disk access requests to reduce delay and improve performance. [2+1+2] CLO4/POb
The disk controller is currently using different disk scheduling algorithms for evaluation. At a certain instant, the disk head is positioned at track 82 and is moving toward the higher-numbered tracks. The pending disk access requests are:

18, 140, 76, 95, 160, 48, 30, 67, 12, 110, 88

The disk contains tracks numbered from 0 to 199.

Apply C-SCAN and C-LOOK disk scheduling algorithms and for each algorithm:

- i. Determine the sequence in which the requests will be processed.
 - ii. Calculate the total head movement.
 - iii. Identify which algorithm provides better performance for this scenario based on the calculated results.
- b. A large online retail company stores customer orders, payment records, and inventory data on a centralized disk storage system. During a flash-sale event, a large number of disk access requests arrive simultaneously. To improve system responsiveness, the operating system administrator wants to evaluate different disk scheduling algorithms and determine which one performs best for the current workload. At a particular moment, the disk head is positioned at track 72 and is moving toward the higher-numbered tracks. The pending disk access requests are: [2+1+2] CLO4/POb

15, 180, 68, 90, 145, 78, 40, 102, 8, 130, 55

The disk tracks are numbered from 0 to 199.

Apply the following disk scheduling algorithms:

- FCFS (First Come First Serve)
- SSTF (Shortest Seek Time First)
- SCAN

For each algorithm:

- i. Determine the order in which the requests are served.
- ii. Show the head movement steps with individual seek distances.
- iii. Calculate the total head movement to determine which algorithm is the best.

QUESTION 2 [10 MARKS]

- a. While editing a 4 GB video file, a user copies the file from an SSD to an external hard drive. Describe how Direct Memory Access (DMA) transfers the data and explain why DMA is preferred even though the CPU could perform the transfer itself. [5] CLO1/POa
- b. A user creates two links to a file: one using a hard link and another using a soft (symbolic) link. Later, the original file is deleted. What happens to each link and explain the reason. [5] CLO1/POa

OR

- a. A student is running a large database application that frequently accesses memory. Explain why a Translation Lookaside Buffer (TLB) is needed in this scenario. [5] CLO1/POa
- b. Several applications such as a web browser, a media player, and a text editor are running simultaneously on a computer. Explain how memory virtualization helps these applications run safely and efficiently by stating the goals of memory virtualization. [5] CLO1/POa

QUESTION 3 [10 MARKS]

- a. A software development company operates a cloud-based project management system where multiple users frequently access different memory pages during execution. Due to limited main memory availability, the operating system must decide which pages to replace whenever a page fault occurs. The system administrator wants to compare different page-replacement algorithms using the same page-reference pattern. Assume the system has 3 page frames available in main memory. The following page-reference string is generated during program execution:
7, 2, 9, 1, 2, 7, 4, 2, 5, 7, 1, 4, 2
Apply the following page-replacement algorithms:
 - FIFO (First-In First-Out)
 - LRU (Least Recently Used)
- [4+3+3] CLO3/POb

- Optimal Page Replacement

For each algorithm:

- Show the frame allocation process step by step.
- Identify page hits and page faults.
- Calculate the total number of page faults to determine which algorithm shows the best performance.

QUESTION 4 [10 MARKS]

- a. A fintech company operates a secure online transaction processing system where multiple processes compete for limited system resources. To avoid deadlock while handling thousands of financial operations, the operating system uses the **Banker's Algorithm** to determine whether the system is in a safe state before allocating additional resources.

The system currently has **five processes (P0-P4)** and **three resource types (A-C)**:

- Resource A = CPU Units
- Resource B = Memory Blocks
- Resource C = I/O Devices

The following tables show the current system state.

Table 1: Allocation Table

Process	A	B	C
P0	1	0	1
P1	2	1	0
P2	1	3	2
P3	1	1	1
P4	0	2	1

Table 2: Maximum Requirement Table

Process	A	B	C
P0	3	2	2
P1	4	2	2
P2	2	5	3
P3	3	3	3
P4	1	4	2

Handwritten notes and calculations:

- work 3 2 2
- need 5 9 4
- 221 < 322
- P0 322
- 212 < 423
- 121 < 633
- 224 < 765
- 141 < 976
- 322
- 221
- 322
- 532
- 664
- 775
- 796

Handwritten numbers: 2, 2, 2, 3

Available resources for process **A**, **B** and **C** are **3**, **2** and **2** respectively.

- i. Apply the **Banker's Algorithm** to determine whether the system is in a safe state.
 - ii. If the system is safe, determine a **safe sequence**.
- b. Consider the previous example in 4(a). Suppose a new request arrives (0,1,1). Will the system still be at safe state? Justify your answer. [5] CLO2/POb

QUESTION 5 [10 MARKS]

- a. A university computer lab maintains a shared file system for students, faculty, and administrators. Recently, several problems have been reported: [2x5] CLO5/POb

- Students accidentally deleted shared project files.
- Faculty members cannot efficiently locate departmental documents.
- Some confidential examination files were accessed by unauthorized users.
- Multiple users editing the same file caused data inconsistency.

Analyze the existing file management situation and propose appropriate file organization and protection mechanisms to solve these problems. Your answer should discuss the followings:

1. Suitable directory structure
2. Appropriate file access methods
3. File sharing and concurrency control mechanisms
4. Security and access protection techniques
5. How these mechanisms improve overall system resource management.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 2025
Program: B.Sc. in CSE
4th year 1st Semester

Course Title: Robotics
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 437
Full Marks: 50

There are Five Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

Cox's Bazar is a popular tourist beach that experiences excessive plastic waste accumulation, especially during weekends. Manual cleaning is slow and labor-intensive. To improve cleanliness and environmental sustainability, engineers propose a robotic garbage collection system that can patrol the beach, detect litter, collect plastic waste, avoid obstacles such as people and rocks, and notify authorities when its storage compartment becomes full.

- a. Explain the purpose and functional requirements of the beach garbage collection robot. Illustrate the corresponding embedded system diagram. [4] CLO1/PLO-a
- b. Write pseudocode for the robot's operation process from startup to waste collection and disposal notification. Explain how the robot uses sensors and decision-making logic to perform its tasks efficiently. Explain the error handling logic in the work flow. [6] CLO2/PLO-e

QUESTION 2 [10 MARKS]

An earthquake at Gazipur damages a multi-floor shopping complex, leaving many people trapped inside. Dust, smoke, and broken infrastructure make evacuation difficult. A rescue robot is designed to autonomously move between damaged sections, identify safe evacuation routes, detect survivors through cameras, and continuously update maps for emergency responders.

- a. Draw the detailed robot control architecture for this rescue robot. [4] CLO3/PLO-c

- b. Do a case study and develop a SLAM based robot control algorithm for this robot. [6] CLO5/PLO-

QUESTION 3 [10 MARKS]

A food packaging company uses a 2-link robotic arm to place snack packets into boxes.

The arm has:

- Link 1 length = 5 cm
- Link 2 length = 4 cm
- Joint angles:
 - $\theta_1 = 40^\circ$
 - $\theta_2 = 30^\circ$
- The robot arm puts a downward force of 25 N.

During operation, the robot occasionally misses the target box position.

- a. Determine the approximate position of the robotic arm's end effector [4] CLO2/PLO-e
- b. The robot places packets slightly away from the target even though the calculations are correct. Suggest one possible sensor-related reason for this issue and explain your answer. [2] CLO2/PLO-e
- c. Which sensor would be most suitable for detecting whether a packet is present before gripping? How to differentiate the packet from the environment. [4] CLO2/PLO-e

OR

An automobile manufacturing plant Mottoman 10 robot manipulator for high-precision welding operations. The robot repeatedly lifts a welding tool and positions it accurately on vehicle frames. Engineers observe overheating in the base motor, missing welding position and slight positioning errors during long operation periods.

The welding tool applies a downward force of 25 N.

The arm links are:

- Link 1 = 0.6 m
- Link 2 = 0.5 m
- Link 3 = 0.4 m

- a. Calculate the DoF of the robotic arm. [4] CLO2/PLO-e
- b. Explain why increasing the number of links in a robot arm improves flexibility but may create mechanical and control challenges. [2] CLO2/PLO-e
- c. Calculate the approximate torque required at the base joint when the arm is fully extended horizontally. [4] CLO2/PLO-e

QUESTION 4 [10 MARKS]

A hospital uses a Baxter Robot to assist surgeons during minimally invasive tumor biopsy procedures. A medical camera mounted above the patient continuously captures images to guide the robot arm toward the target tissue. During testing, slight servo motor inaccuracies caused the surgical needle to deviate from the intended position.

- a. Draw the state diagram of the robot. [4] CLO3/PLO-c
- b. The robotic arm vibrates slightly while holding surgical instruments. Suggest one possible control-system reason for the vibration and discuss one improvement with technical proof that could stabilize the movement. [6] CLO2/PLO-e

QUESTION 5 [10 MARKS]

A powerful earthquake occurred at Patuatuli, trapping civilians inside the building. Fires erupt on several floors due to damaged electrical systems. Thick smoke and blocked stairways make evacuation difficult. A rescue robot is designed to autonomously explore the building, identify safe evacuation paths, and continuously communicate with emergency responders.

- a. Draw the embedded diagram of this robot. [3] CLO1/PLO-a
- b. Which communication protocol will be better here? Provide justification behind your answer. [3] CLO4/PLO-b
- c. Draw the control flow diagram of the robot operation. [4] CLO2/PLO-e

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 2025
Program: B.Sc. in CSE
4th year 1st Semester

Course Title: Machine Learning and Deep Learning
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 413
Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. An insurance company wants to build a Decision Tree model to predict whether a customer will purchase a premium health insurance plan. Based on the past data, the company collected the following dataset shown in **Table 1**. [7] CLO2/
PLOc

Table 1. Dataset for Decision Tree

ID	Age Group	Income Level	Marital Status	Existing Insurance	Buys Product
1	Young	High	Married	No	No
2	Young	High	Single	No	Yes
3	Adult	High	Married	Yes	Yes
4	Adult	Low	Married	Yes	Yes
5	Adult	Low	Single	No	Yes
6	Adult	Low	Married	No	No
7	Adult	Low	Single	Yes	Yes
8	Young	Low	Married	No	No

- i) Using Entropy and Information Gain, determine best feature for the root node.
- ii) Construct the Decision Tree step-by-step using the selected root.
- b. A retail company wants to analyze customer data containing age, annual income, shopping frequency, and product preferences. The company has two different tasks: [3] CLO2/
PLOc
- Task-1: To predict whether a new customer will buy a premium membership based on past customer records where the membership status ("Yes" or "No") is already known.
- Task-2: To discover natural groups of customers with similar purchasing behaviors without having any predefined labels.
- Identify which of the above tasks should use **binary classification** and which should use **clustering**? Explain why?

QUESTION 2 [10 MARKS]

- a. A hospital is evaluating the performance of a machine learning model used to predict whether a patient has a particular disease. After testing the model on 10 patients, the following actual and predicted classifications (shown in **Table 2**) were obtained:

Table 2. Actual and Predicted Values from Test Set

Patient No.	Actual Class	Predicted Class
1	Positive	Positive
2	Positive	Negative
3	Negative	Negative
4	Negative	Positive
5	Positive	Positive
6	Negative	Positive
7	Positive	Positive
8	Positive	Positive
9	Negative	Negative
10	Positive	Negative

[4 CLO4
+2 PLOb
]

- Construct the confusion matrix for the above model.
 - Evaluate the model based on your derived confusion matrix if the model would be better at avoiding false positives or false negatives.
- b. A teacher wants to predict students' final exam marks based on the number of hours they studied. An AI engineer decides to use Linear Regression on the training dataset shown in **Table 3**, to model the relationship between study hours and exam marks. The proposed hypothesis function is: $h_0(x) = \theta_0 + \theta_1 x$ where $\theta_0 = 30$ and $\theta_1 = 10$.

[4 CLO2/
PLOc

Table 3. Student Dataset

Study Hours (x)	Exam Marks (y)
1	40
2	50
3	60
4	70
5	80

Handwritten notes: $\theta_0 = 30$, $\theta_1 = 10$

Calculate the predicted outputs for all the samples of the dataset in **Table 3** using the given hypothesis function.

OR

- a. Consider the dataset provided in **Table 2**.
- Calculate precision and recall.
 - Evaluate the model based on your derived precision and recall if the model should prioritize the precision or recall? Justify your answer.

[4 CLO4/
+ PLOb
2]

- b. Suppose that a food company is analyzing delivery locations in a city to create two regional service zones for its delivery vehicles. Each delivery location is represented as a 2D point, where the coordinates indicate the relative position of the delivery destination. The following delivery locations are recorded:

[4 CLO2/
PLOc

(2,3), (3,5), (5,1), (8,1), (6,9), (10,6)

The company decides to use the K-Means clustering algorithm (with $k = 2$) where cluster centroids, $C_1 = (3,5)$ and $C_2 = (8,1)$ indicates initial positions of the delivery vehicles.

- Assign each location point to the nearest cluster.
- Recalculate the new cluster centroids after assignment.

QUESTION 3 [10 MARKS]

An IT company is developing two SVM-based classifiers for detecting fraudulent transactions. [3] CLO2/
PLOc

- Dataset A: contains customer transactions that are perfectly separated into fraudulent and non-fraudulent classes.
- Dataset B: contains overlapping classes and several noisy or mislabeled transactions.

Identify and explain which approach (*Hard Margin SVM* or *Soft Margin SVM*) is suitable and would perform better in case of each of the datasets.

- b. A hospital is developing an SVM-based medical diagnosis system to classify tumors as either Healthy (+1) or Malignant (-1). Each tumor sample is represented using two numerical features as: Cell size (x_1) and Nucleus irregularity (x_2) as shown in **Table 4**. [7] CLO2/
PLOc

Table 4. Actual and Predicted values from test set

Cell Size (x_1)	Nucleus irregularity (x_2)	Output (y)
1	5	Healthy
2	4	Healthy
3	6	Healthy
4	5	Healthy
5	2	Malignant
6	1	Malignant
7	3	Malignant
8	2	Malignant

After analyzing the dataset, the following support vectors are identified:

- Healthy support vector: (3,5)
- Malignant support vectors: (5,2) and (6,3)

Answer the following:

- Generate the SVM equations using the support vectors.
- Determine the equation of the optimal separating hyperplane (decision boundary).
- Show the visualization of the SVM model.

QUESTION 4 [10 MARKS]

- a. A researcher is training a neural network for handwritten digit classification. During experimentation, they observe that changing the **learning rate** significantly affects the training process. Explain how a very small, and a very large learning rate affect the convergence behavior of neural network training. [3] CLO3/
PLOc

- b. A smart home company is developing a perceptron-based security system to determine whether an alarm should be activated. The system uses three sensor features as Motion detected, Door opened and Window opened. The alarm should activate ($y = 1$) only when multiple features are detected together. The training data is provided in **Table 5**. [2] CLO3/
+ PLOc
2.5
+
2.5
]

Table 5. Training data for smart home system

Motion detected (x_1)	Door opened (x_2)	Window opened (x_3)	Alarm (y)
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	1
1	1	1	1

The parameters of the perceptron are as follows:

- "Step" activation function
 - Initial weights $w_1 = 0.1$, $w_2 = -0.2$, $w_3 = 0.2$
 - Bias, $b = -0.1$ and Learning rate, $\alpha = 0.2$
- Illustrate a perceptron for the given scenario, labeling the inputs, weights, bias, summation, activation function, and output clearly.
 - Compute one full iteration (epoch) of perceptron training using the dataset.
 - Show the detailed weight updates after each training example.

QUESTION 5 [10 MARKS]

- A computer vision engineer is developing a CNN-based security system to detect suspicious objects from surveillance images. As part of the feature extraction stage, the engineer applies a convolution operation followed by max pooling. The CNN uses stride = 1, and no padding. Construct the output of each layer in the following CNN, where a 4×4 image matrix and a 3×3 convolution kernel is given in **Figure 1**. [4] CLO3/PLOc

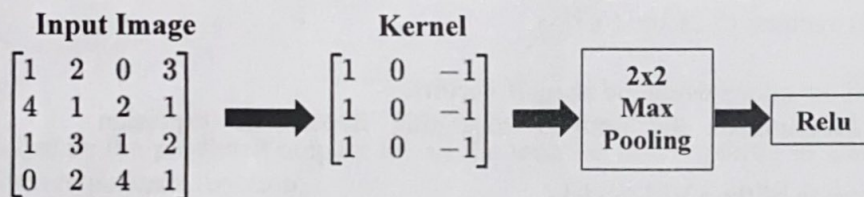


Figure 1. Convolutional Neural Network

- A healthcare company is developing three AI applications: [6] CLO4/PLOb
 - App1: A system that detects pneumonia from chest X-ray images.
 - App2: A system that analyzes a patient's heartbeat readings over time to detect irregular patterns.
 - App3: A system that predicts whether a patient has diabetes using features such as age, blood pressure, glucose level, and BMI.
 - Identify the most suitable neural network architecture for each AI applications from the following:
 - Regular Feedforward Neural Network (ANN)
 - Convolutional Neural Network (CNN)
 - Recurrent Neural Network (RNN)
 - Evaluate your selected model for each application based on the nature of the input data and the problem being solved.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 2025
Program: B.Sc. in CSE
4th year 1st Semester

Course Title: Numerical and Mathematical Analysis for Engineers
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 405
Full Marks: 50

There are Five Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Explain the three types of Runge-Kutta 2nd Order Methods. How are they derived from the general Runge-Kutta 2nd Order equation? [3] CLO2/
PLOa
- b. A material engineering company is testing the cooling behavior of a newly manufactured high-temperature steel sphere used in an industrial furnace. Immediately after production, the sphere is removed from the furnace at a temperature of 1200 K and exposed to the surrounding air, which is maintained at an ambient temperature of 300 K. The engineers assume that the sphere loses heat only through thermal radiation, and the temperature variation of the sphere is governed by the following differential equation: [7] CLO3/
PLOa

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} (\theta^4 - 81 \times 10^8), \theta(0) = 1200 K$$

where θ represents the temperature (K) and t is the time (s).

Using a step size of $h=120$ seconds, determine the temperature of the sphere at $t=240$ seconds by applying Heun's Method, and Midpoint Method (Runge-Kutta 2nd Order methods) and compare the numerical results by calculating the differences between the temperatures obtained from the ~~three~~ methods.

QUESTION 2 [10 MARKS]

- a. Explain the algorithm of finding the root of any equation using Newton-Raphson method. [3] CLO2/
PLOa
- b. A renewable energy company is designing a control system for a wind turbine, where a nonlinear mathematical model must be solved to determine the optimal operating parameter. The parameter x satisfies the equation I [7] CLO3/
PLOa

$$f(x) = x^3 - x - 1 \quad (\text{Equation I})$$

To obtain an approximate solution, the engineering team decides to use the Newton-Raphson method with an initial guess of $x_0 = 1.5$. Apply three iterations of the Newton-Raphson method to estimate the root of the equation and calculate the absolute relative approximate error at the end of each iteration.

QUESTION 3 [10 MARKS]

- a. A game development company is designing a circular elimination game in which players are removed one by one according to a fixed rule until only one player remains. The position of the surviving player is represented by $J(n)$, where n is the total number of players. Determine the minimum five values of n for which the last player in the original arrangement survives the elimination process. Justify your answer using the appropriate formula. [5] CLO4/
PLOb
- b. A software company is developing a circular task-scheduling system in which tasks are repeatedly removed according to a specific elimination rule until only one task remains active. The position of the surviving task is represented by $J(n)$, where n is the number of tasks initially present. The system follows the recurrence relations: [5] CLO4/
PLOb

$$J(2n) = 2 \cdot J(n) - 1$$

$$J(2n+1) = 2 \cdot J(n) + 1$$

As a system analyst, use the above recurrence relations to determine the surviving position when there are 230 tasks in the system. Calculate $J(230)$ by repeatedly applying the appropriate recurrence relation(s), showing all intermediate steps and the final result.

OR

- a. You are analyzing an optimal solution for 3 disks in the Tower of Hanoi problem. The moves are listed below where peg 1, peg 2, peg 3 are represented by Source, Auxiliary and Target. The moves are: [5] CLO4/
PLOb
1. Source $\rightarrow$ Target
 2. Source $\rightarrow$ Auxiliary
 3. Target $\rightarrow$ Auxiliary
 4. Source $\rightarrow$ Target
 5. Auxiliary $\rightarrow$ Source
 6. Auxiliary $\rightarrow$ Target
 7. Source $\rightarrow$ Target
- i) After move 3, demonstrate which disks are on which peg?
- ii) Given, move 5 is "Auxiliary $\rightarrow$ Source". Identify why this move is Necessary?
- b. In the Tower of Hanoi puzzle, n disks are stacked on the first peg in order of decreasing size (largest at the bottom, smallest at the top). The goal is to move [5] CLO4/
PLOb

all disks from the first peg to the third peg using the second peg as an auxiliary, following these rules:

- i) Only one disk can be moved at a time.
- ii) A larger disk can never be placed on top of a smaller disk.

Let $T(n)$ be the minimum number of moves required to solve the Tower of Hanoi puzzle with n disks. It is claimed that: $T(n) = 2^n - 1$ for all integers $n \geq 1$. Using mathematical induction, prove that $T(n) = 2^n - 1$ for all $n \geq 1$.

QUESTION 4 [10 MARKS]

- a. A quality control engineer is testing a sensor device that can produce one of two outcomes during a diagnostic check: Success (Head) or Failure (Tail). To evaluate the reliability of the device, the engineer performs three independent tests. Let the random variable X represent the number of successful tests (Heads) obtained in the three trials. [1+2 CLO4/
+2] PLOb 7
- i. Analyze all possible outcomes of the three diagnostic tests and determine the possible values of the random variable X .
 - ii. Construct the probability distribution table of X by classifying the outcomes according to the number of successes obtained.
 - iii. Using the probability distribution, determine: $\frac{\sum x \cdot P(x)}{n}$
 - $P(X=2)$
 - $P(X \geq 1)$
- b) A gaming company is testing a new board game that uses two fair six-sided dice. During each trial, both dice are rolled simultaneously, and the player's movement is determined by the sum of the numbers shown on the two dice. Let the random variable Y represent the sum obtained from a single roll of the two dice. [1+2 CLO4/
+2] PLOb
- i. Analyze all possible outcomes when two fair dice are rolled and determine the possible values of the random variable Y .
 - ii. Construct the probability distribution table of Y .
 - iii. Using the probability distribution, determine
 - $P(Y=7)$
 - $P(Y>9)$
 - $P(Y \leq 4)$

QUESTION 5 [10 MARKS]

- a. A university is developing a data-driven student performance prediction system to identify students who may need additional academic support. Historical data collected from previous students include four attributes: Study Hours, Attendance, Assignment Submission, and Previous GPA, along with their final Performance in Class (A, B, or C). The dataset is shown below.

[10] CLO4/
PLOb

Study Hours	Attendance	Assignment Submission	Previous GPA	Performance in Class
High	High	Yes	High	A
High	Medium	Yes	High	A
Medium	High	Yes	High	A
High	High	Yes	Medium	A
Medium	Medium	Yes	High	A
High	High	No	High	A
Medium	High	Yes	Medium	A
High	Medium	Yes	Medium	B
Medium	Medium	Yes	Medium	B
Low	High	Yes	Medium	B
Medium	Low	Yes	Medium	B
Medium	Medium	No	Medium	B
High	Low	Yes	Low	B
Low	Medium	Yes	Medium	B
Medium	High	No	Low	B
Low	Low	No	Low	C
Low	Medium	No	Low	C
Medium	Low	No	Low	C
Low	Low	Yes	Low	C
Low	Medium	No	Medium	C
Medium	Low	No	Medium	C
Low	High	No	Low	C
Low	Low	No	Medium	C
Medium	Medium	No	Low	C

A new student has recently enrolled in the university. Based on the student's academic profile (Study Hours = Medium, Attendance = High, Assignment Submission = Yes, Previous GPA = Medium), analyze an appropriate classification technique to predict the student's Performance in Class. Show the necessary calculations and justify the predicted class label.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 2025
Program: B.Sc. in CSE
4th year 1st Semester

Course Title: Industry and Operational Management
Time: 3:00 hours

Credit Hour: 3.00

Course Code: IMG 401
Full Marks: 50

There are **Five Questions**. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

A technology company launches an online video streaming platform that serves millions of users daily. To stay ahead of competitors, the company focuses on releasing new features very quickly and minimizing operational costs. As a result, some testing and monitoring activities are reduced. Within a few days of launch, users begin experiencing frequent buffering, system crashes, and service downtime during peak hours. The company also struggles to handle the sudden increase in user traffic.

- | | | |
|---|-------|----------------|
| a. Define operations management and list the core elements of operations management. | [1+2] | CLO1/
PLO a |
| b. Explain important operational focuses and failure impact of the industry from an operations management perspective. | [4] | CLO1/
PLO a |
| c. Using the Cost-Speed-Quality trade-off triangle, explain the factors the company prioritized and the negative consequences that occurred because of this decision. | [3] | CLO1/
PLO a |

QUESTION 2 [10 MARKS]

A rapidly growing software company recently changed its organizational structure from a simple centralized structure to a matrix structure so that employees from different departments such as engineering, marketing, and product design can work together on multiple projects simultaneously. However, many employees have become confused because they now report to both functional managers and project managers. At the same time, the company noticed that employee motivation and productivity are decreasing. Employees feel that their extra efforts are not being properly recognized, and some believe that rewards and promotions are unfair compared to others in the organization.

- | | | |
|--|-------|----------------|
| a. Examine the type of organizational structure used by the company and analyze the disadvantages of this structure. | [2+3] | CLO2/
PLO i |
|--|-------|----------------|

- b. Analyze the employees' feeling of unfair rewards and reduced motivation using an appropriate motivation theory.

[5] CLO2/
PLO i

QUESTION 3 [10 MARKS]

A computer hardware manufacturing company recorded the monthly demand (in units) for a newly launched graphics card over the past 12 months:

Table 1. Monthly Graphics Card Demand

Month	Demand (Units)	Month	Demand (Units)
January	420	July	470
F ₂ February	435	August	488
March	428	September	495
F ₄ April	450	October	510
F ₅ May	462	November	505
June	455	December	525

- a. Using a 3-month weighted moving average, compute the forecast up to next year's January, where the weights for the most recent month is 0.55, the second most recent month is 0.3, and the third most recent month is 0.15
- b. Using Exponential Smoothing, calculate the forecast up to next year's January, given: Smoothing constant, $\alpha = 0.25$
- c. Analyze both the forecasting techniques using MAD and MAPE from April to December.

[3] CLO3/
PLO k

[3] CLO3/
PLO k

[4] CLO3/
PLO k

QUESTION 4 [10 MARKS]

A medical device manufacturing company launched a new patient monitoring system for hospitals. To reduce production time and operational costs, management shortened inspection procedures and skipped several quality assurance reviews before shipment. Within a few months, hospitals began reporting device malfunctions, inaccurate readings, and frequent maintenance issues. Customer complaints increased rapidly, and the company faced warranty claims, emergency recalls, and reputational damage. To address the situation, management formed a cross-functional quality improvement team to investigate the root causes and redesign the operational process.

- a. Analyze the company's quality management failure using the Cost of Quality (CoQ) concept and explain how prevention, appraisal, and failure costs are reflected in the scenario.
- b. Analyze how quality management tools and process improvement frameworks could help the company improve product reliability and operational performance.

[2+3] CLO3/
PLO k

[5] CLO3/
PLO k

OR

A software company is developing a large e-commerce platform with multiple interconnected activities such as database setup, payment gateway integration, frontend development, testing, and deployment. During project execution, management discovers that several developers are assigned to multiple critical activities at the same time, causing resource conflicts and delays. In addition, unexpected client feature requests increase the project scope, making the original schedule difficult to maintain.

- a. Analyze how resource leveling and resource smoothing could help the company manage resource conflicts and schedule delays. [5] CLO3/
PLO k
- b. Analyze the scheduling challenges faced by the company and explain how these challenges could affect project completion and operational performance. [2+3] CLO3/
PLO k

QUESTION 5 [10 MARKS]

A computer hardware manufacturing company is planning to purchase microcontroller chips from two suppliers under different quantity discount schedules. The monthly demand for the chips is 850 units, ordering cost is \$75 per order, and annual holding cost is \$6 per unit. The management wants to determine the most cost-effective purchasing option considering ordering, holding, and purchasing costs.

Table 2. Price List

Supplier A		Supplier B	
Quantity Range	Unit Price (\$)	Quantity Range	Unit Price (\$)
1 – 299	22.6	1 – 249	22.8
300 – 599	22.1	250 – 549	22.3
600 – 899	21.8	550 – 849	21.9
900 – 1499	21.3	850 – 1349	21.5
1500+	20.9	1350+	21

- a. Compute the optimal order quantity using the EOQ model. [2] CLO3/
PLO k
- b. Analyze the quantity discount alternatives to determine the optimal supplier and order quantity by calculating the total annual cost. [8] CLO3/
PLO k



University of Asia Pacific
Department of Computer Science and Engineering
Semester Final Exam Routine Fall 2025

Date	Time	1 st Year 1 st Semester	1 st Year 2 nd Semester	2 nd Year 1 st Semester	2 nd Year 2 nd Semester	3 rd Year 1 st Semester	3 rd Year 2 nd Semester	4 th Year 1 st Semester	4 th Year 2 nd Semester
07.06.2026 Sunday	10:00AM~01:00PM	CSE 101			CSE 209 *CSE 209 **CSE 335	CSE 305		CSE 401 *CSE 405 **CSE 303	
08.06.2026 Monday	02:00PM~05:00PM		CSE 105 *CSE 105	CSE 205 *CSE 407			CSE 311 *CSE 303		CSE 429 **CSE 403
09.06.2026 Tuesday	10:00AM~01:00PM	HSS 111 *HSS 111 (A)			CSE 211 *CSE 211 **CSE 321	CSE 301 *CSE 309		CSE 437 *CSE 315	
10.06.2026 Wednesday	02:00PM~05:00PM		CSE 103 *CSE 103	CSE 201 *CSE 203 **CSE 203			CSE 313 *CSE 321		CSE 425 **CSE 431
11.06.2026 Thursday	10:00AM~01:00PM	MTH 101 *MTH 101			CSE 207 *CSE 207	CSE 303 *CSE 317 *CSE 313		CSE 413 *CSE 427	
13.06.2026 Saturday	02:00PM~05:00PM		MTH 103 *MTH 201 *MTH 205 **MTH 103	CSE 203 *CSE 205			CSE 315 *CSE 311 *CSE 319		CSE 457
14.06.2026 Sunday	10:00AM~01:00PM	ENG 101 *HSS 301			MTH 203 *MTH 203 **MTH 203	BIE 301		CSE 405 *CSE 401	
15.06.2026 Monday	02:00PM~05:00PM		CHEM 111	MTH 201 *MTH 103			CSE 317		BUS 401
16.06.2026 Tuesday	10:00AM~01:00PM	PHY 101 *PHY 101 **PHY 101			ECN 201 *ECN 201 **ECN 201	BUS 301		IMG 401 **IMG 201	
17.06.2026 Wednesday	02:00PM~05:00PM		EEE 101 *EEE 121 **ECE 303	EEE 201 *EEE 221 **ECE 301			PHY 301 *CSE 307		
18.06.2026 Thursday	10:00AM~01:00PM			**ACN 201		CSE 307		*CSE 403	

Note: 1st Year 1st Semester to 4th Year 1st Semester OBE Syllabus. 4th Year 2nd Semester 2016-2022 Syllabus. 2016-2022 Syllabus: Applicable from Fall-16 to Spring-22(*). 2011-2015 Syllabus: Applicable up to Spring Spring-16(**). Course marked in black box indicate self-study courses.

Prepared by:

Md. Abdul Mohtashim
Md. Abdul Mohtashim
Laboratory Computer Lab. Assistant
Department of CSE
University of Asia Pacific

Checked by:

DAO/AAO
Ass. Administrative Officer
CSE, UAP
University of Asia Pacific

A S Zafarullah
A S Zafarullah
Assistant Professor
Department of CSE
University of Asia Pacific

Dr. Shah Murtaza Rashid Al Masud
Dr. Shah Murtaza Rashid Al Masud
Professor & Head
Department of CSE
University of Asia Pacific